MILITARY SPECIFICATION

BEACON, RADIO AN/APN-69

1. SCOPE

1.1 This specification covers the general requirements of one type of radio beacon designated as Beacon, Radio AN/APN-69.

1.1.1 Beacon, Radio AN/APN-69 functions as a beacon when properly interrogated by suitable radar sets as specified in Specification MIL-R-5582. The beacon transmitter output is a selectively coded signal permitting different beacons to be independently identified by interrogating radars.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-E-1	Electron Tubes
MIL-T-27	Transformers and Inductors (Audio, Power, and Pulse)
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant, For The Treatment of Communications, Electronic, And Associated Electrical Equipment
JAN-J-641	Jacks, Telephone
MIL-C-3608	Connectors, "BNC," for Radio Frequency Cables
MIL-E-4682	Electron Tubes, Choice and Application Of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-W-5274	Wire, Electrical, Insulated, Aircraft
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-R-5582	Radar; X-Band, Airborne - Standard Beacon Provisions
MIL-R-5757	Relays, Armature (For Electronic and Communication Equipment)
MIL-I-6181	Interference Limits, Tests and Design Requirements Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted

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MIL-P-7094	Plastic Parts, Aircraft Exterior, General Requirements and Tests For Rain-Erosion Protection Of
MIL-E-7894	Electric Power, Aircraft, Characteristics Of
MIL-M-7969	Motors, Alternating Current, 400-Cycle, 115/200 Volt System, Aircraft, General Specification For
MIL-T-8212	Tube, Electron, Type USAF 6233
MIL-F-15160	Fuses; Instrument, Power and Telephone
MIL-E-17555	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing, and Marking Of

STANDARDS

FEDERAL

FED-STD-5	Standard Guides for Preparation of Item Descriptions By Government Suppliers

MILITARY

MIL-STD-16	Electrical and Electronic Reference Designations
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS24123	Plate, Identification
MS25043	Cap-Electrical Connector, Receptacle, Dust
MS90058	Flanges, Waveguide, Choke, UG-40A/U and UG-136A
MS91236	Connector, Plug, Electrical, Series N, UG-21D/U

AIR FORCE-NAVY AERONAUTICAL

AN3102	Connectors - Electrical, Receptacles, Box Mounting

DRAWINGS

AIR FORCE

51C13643	Test Lead Set, CX-1331A/U, Assembly
53C12931	Receiver-Transmitter, Radio RT-204( )/APN-69, Installation of HD-135( )/APN-69 Centrifugal Fan, MT-824( )/APN-69 Mounting, On
53C12932	Control, Radio Beacon C-812( )/APN-69, Outline Dimensions Of
53C12933	Antenna AS-521( )/APN-69, Outline Dimensions Of
53B12934	Coupler, Directional CU-341( )/U, Outline Dimensions Of
53C12935	Wiring Diagram, AN/APN-69( ) Radio Beacon, Interconnecting

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NAVY

RE49F506 Shorting Plug Army-Navy Type CW-159/U For Use With Type BNC Connectors

PUBLICATIONS

AIR FORCE-NAVY AERONAUTICAL BULLETIN

143 Specifications and Standards; Use Of

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

AMERICAN INSTITUTE OF ELECTRICAL ENGINEERS

STD-NO- 1 General Principals Upon Which Temperature Limits Are Based In The Rating of Electrical Machines And Apparatus

(Copies of AIEE publication may be obtained from the American Institute of Electrical Engineers, 33 West 39th Street, New York, New York)

3. REQUIREMENTS

3.1 Preproduction sample.- This specification requires the submission of preproduction samples for preproduction testing.

3.2 Components.- One complete Beacon, Radio AN/APN-69 shall consist of the following major components:

Receiver-Transmitter, Radio	RT-204( )/APN-69	1 each
Control, Radio Beacon	C-812( )/APN-69	1 each
Antenna	AS-521( )/APN-69	1 each
Mounting	MT-824( )/APN-69	2 each
Fan, Vaneaxial	HD-135( )/APN-69	1 each
Coupler, Directional	CU-341( )/APN-69	1 each

3.2.1 Removal of parentheses.- In the type designations above, which include parentheses, the parentheses will be replaced by another symbol. This information will be furnished to the contractor after award of the contract upon application by the contractor to the procuring activity.

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3.3 General.- The requirements of Specification MIL-E-5400 are applicable as requirements for this specification, with the following additions or exceptions to the paragraphs of Specification MIL-E-5400 as identified by similarity of paragraph headings. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.3.1 Approval of parts and materials

3.3.1.1 The sample parts and quantities required shall be specified by the procuring activity after the preliminary list of electronic parts has been submitted and reviewed. The contractor shall test the specified nonstandard parts and materials prior to the submission of the items to the procuring activity for approval. The tests conducted shall be all the appropriate tests of the specification applicable to the particular item as determined by ANA Bulletin 143. The results of the tests shall be reported to the procuring activity, in duplicate, using the nonstandard part data sheet contained in Specification MIL-E-5400 for each nonstandard part. Test data and comments, item 12, of the data sheet shall contain a. the complete detailed test procedure used and b. test data processed to a form which yields direct comparison between the requirements and the test results. The sample nonstandard parts on which the contractor conducted his tests shall be submitted to the procuring activity together with nonstandard part data sheets. The description of part shall adequately describe the part as indicated in the following paragraph.

3.3.1.2 Preliminary list of electronic parts.- The preliminary list of electronic parts shall indicate all component detail parts affecting electrical characteristics. The list shall be used as a check list for determining compliance with parts, material, and process requirements of Specification MIL-E-5400. The preliminary list of electronic parts shall contain accurate identifying information in accordance with Standard MIL-STD-16 and FED-STD-5, including the following:

a. Government order number (file number) or contract number.
b. Prime contractor's name and part number
c. Manufacturer's name, catalog number and MIL type designation
d. An accurate description of the part or material, including the following:

   (1) Number of the standard specification (MIL, AN, RTMA, et cetera)
   (2) Electrical and mechanical ratings
   (3) Special features of construction
   (4) Materials used (type, grade, et cetera)
   (5) Type of insulation, dielectric, et cetera
   (6) Type of finish or covering
   (7) Details of spring materials, contacts, et cetera
   (8) Details of tropicalization process

e. Each page of the list shall be numbered and dated

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3.1.3 Final approval of the equipment required by this specification shall be contingent upon approval of all components and parts by the procuring activity.

3.3.2 Connectors

3.3.2.1 Connectors for voltmeter or milliammeter measurements.- Connectors for voltmeter or milliammeter measurements other than PJ-055B in accordance with Specification JAN-J-641 shall accommodate test prods in accordance with the requirements of Drawing 51C13643.

3.3.2.2 Connectors wired in.- Soldered electrical connectors shall not be painted nor fungus treated. All rosin deposits, however, shall be removed.

3.3.3 Pilot lamps.- Pilot lamps shall be provided with brilliance controls.

3.3.4 Finishes.- Aluminum parts or chassis which need to be bonded or grounded shall not be lacquer finished on surfaces which provide bonding and grounding.

3.3.4.1 Standard parts, cadmium plated in accordance with Specification QQ-P-416 and supplied only with type I cadmium plating shall not be used unless given the type II treatment (or type III, if painted) before assembly into equipment, or unless used within pressurized or sealed space not exposed to the deteriorating effects of moist air.

3.3.5 Fuse posts.- The hole in fuse posts shall permit insertion of a test prod conforming to Drawing 51C13643.

3.3.6 Marking and identification

3.3.6.1 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.3.6.2 The weight of Receiver-Transmitter RT-204/APN-69 shall be marked on the back, the top, and the side of the case providing this weight exceeds 50 pounds.

3.3.6.3 Nameplates.- Nameplates shall conform as closely as practicable to Standard MS24123.

3.3.7 Meters.- Only ruggedized meters shall be acceptable.

3.3.8 Motors (ac).- Motors operating from ac shall comply with the requirements of Specification MIL-M-7969 insofar as the requirements do not conflict with those herein.

3.3.9 Relays.- Relays shall be hermetically sealed, shall be in accordance with Specification MIL-R-5757, shall be shock tested by Shock Method II, seal tested by Seal Test Method II, and shall be subject to approval by the procuring activity.

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3.3.10 Transformers and inductors (audio and power).- Transformers and inductors shall be in accordance with Specification MIL-T-27 except as follows:

3.3.10.1 After voltage surges, at sea level and at pressure of 3.4 inches of mercury, corona shall be completely extinguished when the input returns to the maximum operating value specified in Specification MIL-E-7894. The humidity test for all transformers and inductors shall be as specified by Specification MIL-T-27 except that the period of exposure to freezing temperature shall not apply. Polarizing voltage shall be applied during humidity testing only to types which normally operate with a dc voltage relative to ground. Vibration during the humidity test shall not be required, but prior to the humidity test, transformers and inductors shall be vibrated for 30 minutes in each of three mutually perpendicular planes at a total excursion of 0.03 inch over a frequency range of 10 to 55 cps. This range of 10 to 55 cps and back to 10 cps shall be traversed in approximately 1 minute. At least one sample of sealed types shall be opened and inspected for compliance with the core and coil mounting requirements. The contractor's report of tests shall state the results of this inspection.

3.3.10.2 Insulation temperature.- The insulation temperature shall be within the maximum safe continuous operating temperature limits for the class of insulation employed to provide the life required by this specification. The temperature for class A insulation shall not exceed 125°C. (257°F); for class B insulation, temperature shall not exceed 160°C. (320°F) and for class H insulation shall not exceed 190°C (374°F). Classes A, B, and H are defined in AIEE Standard 1.

3.3.10.3 Rating.- Transformers and inductors shall be so designed that when installed they shall reliably and safely handle the required power, peak voltage, and the specified life, without degradation in their own performance or that of the equipment in which they are employed throughout the life of the equipment.

3.3.10.4 Size and weight.- The size and weight of transformers and inductors shall be as small as practicable, consistent with required performance and life. Every effort shall be made to use materials of light weight and to employ methods of design and construction which effect minimum size and weight. High temperature rise is permitted when size and weight savings can be effected, provided dependability, performance, efficiency, and required life are obtained. The optimum grades of core material shall be used to the maximum extent justifiable for the particular application.

3.3.10.5 Open type.- Open type construction if used for low voltage "power transformer and inductor families" defined in Specification MIL-T-27 shall conform to the following requirements.

3.3.10.5.1 Temperature and immersion cycling.- During the last hot cycle of the temperature cycling of Specification MIL-T-27, without salt water immersion, that is step 5, open type units shall be subjected to an overvoltage, condition 1, test

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for 1 minute at 125 percent of rated voltage and frequency, and subjected to a dielectric strength test at 75 percent of the specified test voltage for 1 minute. After the temperature cycling followed by immersion in tap water for 24 hours at 20° to 30°C, the insulation resistance of open-type transformers and inductors shall not fall below 50 megohms between windings and core, and between windings. The time limit specified for the temperature cycles shall apply after the samples are stabilized at each extreme temperature. The sealing test of Specification MIL-T-27 shall not apply.

3.3.10.5.2 Moisture resistance.- The minimum insulation resistance immediately after the moisture resistance test shall not be less than 25 megohms. After a 24-hour drying period at overvoltage, condition 2, insulation resistance shall not fall below 50 megohms between windings and core, and between windings.

3.3.10.5.2.1 It is intended that the insulation resistance of the transformers and inductors be as specified in 3.3.10.5.2 immediately after the moisture resistance test. If this resistance is not obtained the transformers and inductors may be dried with their own internally generated heat at normal room conditions but protected from direct drafts and sources of heat. The performance shall be measured at periods not greater than 30 minutes until specified performance is obtained or until failure is definitely established. Complete data of this procedure shall be taken, evaluated and submitted to the procuring activity. Based upon the information submitted and other pertinent factors, the procuring activity shall decide whether the transformers and inductors are satisfactory for the intended use so far as moisture resistance is concerned.

3.3.10.5.3 Sampling.- Failure of an open-type unit on sampling tests for temperature rise, cycling and immersion, and overvoltage, condition 2, shall require testing each unit in the lot to determine whether the unit is acceptable. The samples tested for temperature rise, cycling and immersion, and overvoltage shall be acceptable for shipment on the order if they meet all requirements.

3.3.10.6 Potting.- Unless proof of improved heat dissipating ability can be supplied, sealed transformers and inductors having a temperature rise greater than approximately 15 degrees C shall not be filled with potting compound.

3.3.10.7 Shielding.- Power transformers shall be electrostatically shielded.

3.3.11 Tubes (electron).- The choice of tubes and their applications shall be in accordance with Specification MIL-E-4682.

3.3.12 Wire (hook-up).- Wire in accordance with Specification MIL-W-5274 shall be used where high insulation resistance is dictated by circuit design.

3.3.13 Fungus treatment.- Fungus inert materials and parts shall be used to the greatest extent possible. Overall spray or brushing shall not be used. Any materials or components which must be used that are not in themselves inert to fungus growth shall be treated individually unless they are within hermetically sealed cases. Fungistatic coating material, if used, shall be type I in accordance with Specification MIL-V-173.

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3.3.14 Life.- The equipment shall be capable of service-free operation for the first 200 hours of the life test specified herein. Parts requiring servicing or replacement at the end of the 200-hour interval shall be specified by the contractor.

3.3.14.1 The equipment shall have a minimum total operating life of 1000 hours with reasonable servicing and replacement of parts. Parts requiring servicing or replacement within this interval and the life of such parts shall be specified by the contractor.

3.3.15 Orientation.- Normal operation of the radio beacon shall be obtained when the receiver-transmitter is mounted in a horizontal position, but no damage shall result when the equipment is temporarily inverted during flight.

3.3.16 Overload protection.- Fuses shall be style F05 or F06 in accordance with Specification MIL-F-15160, Littlefuse types 411, 413, 414 or approved equal.

3.3.17 Service conditions (electrical)

3.3.17.1 The equipment shall perform as specified herein when operated during the cruise-combat conditions from a type III system in accordance with Specification MIL-E-7894, 108 to 121 volts, phase to neutral, except that frequency shall be 380 to 1000 cps.

3.3.17.2 Where dc power is required, the equipment shall perform as specified herein over the range of 25 to 29 volts, and the power shall be used only for such items as relays, magnetic devices, and motors for adjustable speed drives. Where only a small load exists, the dc source shall not be used, and the load may be supplied by individual rectifiers where this will result in weight reduction.

3.3.17.3 Power ratings.- The power required by the equipment shall not exceed the following:

a. 625 volt-amperes from the 380 to 1000 cps source
b. 25 watts from the dc source

3.3.17.4 Power source transients.- The equipment shall be so designed that no damage nor degradation in performance shall result from the transient test specified herein.

3.3.17.5 Power input connector.- The power input connector shall be AN3102C-24-20P.

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3.3.18 Service conditions mechanical

3.3.18.1 Vibration.- The radio beacon shall be capable of withstanding the vibration test specified herein without degradation in performance nor damage to the equipment.

3.3.18.2 Shock.- The radio beacon shall be capable of withstanding the shock test specified herein without impairing the performance of the equipment.

3.3.19 Stability.- The equipment shall operate with specified performance for at least 48 hours, continuously or intermittently without adjustment of any controls.

3.4 Relation of standard sample to this specification.- The standard sample radio beacon furnished as provided in 3.6 shall serve as a standard sample to demonstrate the minimum acceptable standard of materials, design, workmanship, and performance not specified herein. The standard sample shall also serve to determine the mechanical and electrical interchangeability requirements of Specification MIL-E-5400. Provisions for mounting the radio beacon shall be such that physically existing brackets, racks, connector position, cable routing, et cetera, may be used as indicated in applicable installation drawings which may be obtained upon application to the procuring activity.

3.5 Design

3.5.1 Receiver-Transmitter, Radio RT-204( )/APN-69.- Receiver-Transmitter, Radio RT-204( )/APN-69 shall incorporate all necessary circuits and assemblies to satisfy the performance requirements stated herein and shall contain the following major assemblies:

a. Microwave unit consisting of a duplexer, mixer, receiver local oscillator, discriminator, and magnetron
b. Receiver chassis consisting of intermediate frequency amplifier, detector, dc restorer, and video amplifier stages
c. Pulse control chassis consisting of a pulse-length discriminator, blanking circuit, aural monitor, suppressor circuit, test oscillator, duty cycle limiter, and afc aiding circuits
d. AFC chassis consisting of automatic-frequency control circuitry for the magnetron and local oscillator
e. Low-voltage power supply and coder chassis consisting of a magnetostriction-type coder and a power supply for low voltages required by complete equipment
f. Modulator and high-voltage power supply chassis
g. Test panel containing some of the controls and test points as specified in 3.5.1.5.1 and 3.5.1.6

The receiver chassis, pulse-control chassis, afc chassis, low-voltage power supply and coder chassis, and test panel shall be removable for maintenance purposes without the necessity of unsoldering wires or requiring major disassembly procedures.

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3.5.1.1 Dimensions.- The dimensions of the receiver-transmitter shall be in accordance with Drawing 53C12931. The receiver-transmitter shall be as light as practical consistent with the required performance and life. Every effort shall be made to use materials of light weight and employ methods of design and construction which effect minimum weight.

3.5.1.2 Weight.- The weight of the receiver-transmitter shall not exceed 75 pounds. The case shall be capable of being pressurized. The closed end of the case shall be capable of supporting the entire weight of the receiver-transmitter without permanent deformation.

3.5.1.3 Cooling system.- The cooling system for the receiver-transmitter shall limit the temperature rise of the unit to a maximum of 30 degrees C. External and internal blowers or centrifugal fans shall be used. One internal blower or fan shall be used to provide cooling for the magnetron and shall limit the magnetron anode temperature to a maximum of 125 degrees C. All motors shall be permanently lubricated and thermostatically controlled and shall operate from a 115 volt, 380 to 1000 cps ac source.

3.5.1.3.1 Cooling system control.- Snap disc-type thermostats shall be used to energize internal and external blower or fan motors. The magnetron blower motor and the internal blower motor shall be energized when the internal temperature is plus 10 degrees C or higher. The external blower shall be energized when the internal temperature is plus 50 degrees C or higher.

3.5.1.4 Pressurization.- The receiver-transmitter unit shall withstand an internal pressure of 20 pounds per square inch. The pressure leakage rate of the receiver-transmitter shall not exceed 1 pound per square inch in 5 hours from a gage pressure of 20 pounds per square inch over a temperature range of minus 55 degrees C to 71 degrees C. The receiver-transmitter shall be equipped with an air valve and a plug fitting to permit flushing the case with dry air in lieu of using a desiccant.

3.5.1.5 Panel connectors.- Appropriate multiple-contact AN type receptacles for power, Test Set AN/APM-67, and control circuits, standard waveguide flange for connection with the directional coupler and waveguide transmission line, phone jack for the aural monitor circuit, and type BNC receptacles in accordance with Specification MIL-C-3608 for pulse circuits and panel test points shall be provided on the front panel of the receiver-transmitter. All receptacles shall be appropriately marked and those receptacles not normally used, such as test points, shall be provided with a cap and captive chain in accordance with Standard MS25043. Major component interconnection shall be in accordance with Drawing 53C12935.

3.5.1.5.1 External connectors and test points.- The following connectors and test points shall be provided at the front panel of the receiver-transmitter, as shown

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on Drawing 53C12931, and shall be identified by the titles given in the following paragraphs:

a. J-401.- A pressurized multiple-contact receptacle, AN3102C-24-20P, shall provide power input connection.

b. J-402.- A pressurized multiple-contact receptacle, AN3102C-14S-5S, equipped with a cap and captive chain assembly, shall provide connection for Test Set AN/APM-67

c. J-403.- A pressurized multiple-contact receptacle, AN3102C-22-14S, shall provide connection for Control, Radio Beacon C-812( ) /APN-69

d. J-502.- SUPP. A BNC receptacle, type UG-290/U, equipped with a cap and captive chain assembly CW-123/U, both in accordance with Specification MIL-C-3608, shall provide connection for the radar suppressor pulse

e. XTAL CUR.- A BNC receptacle, type UG-254A/U in accordance with Specification MIL-C-3608 equipped with a shorting plug CW-159/U, in accordance with Drawing RE49F506, shall permit measurement of the receiver crystal current.

f. MAG CUR.- A BNC receptacle, type UG-290/U, in accordance with Specification MIL-C-3608, located on the test panel and equipped with a cap and captive chain assembly CW-123/U shall permit measurement of the magnetron anode current

g. AURAL MON.- A telephone jack, JJ-034, in accordance with Specification JAN-J-641, located on the test panel and equipped with a spring-loaded cover, shall provide connection to a headset for aural-monitoring purposes

h. MONITOR.- A BNC receptacle, type UG-254A/U, equipped with a cap and captive chain assembly CW-123/U shall provide connection for oscilloscope viewing of the modulator voltage pulses

i. TEST TRIG.- A BNC receptacle, type UG-290/U, located on the test panel and equipped with a cap and captive chain assembly CW-123/U, shall permit external triggering of the beacon and provide a trigger pulse to synchronize an oscilloscope

3.5.1.6 External controls.- The following controls shall be available at the front panel of the receiver-transmitter and shall be identified by the titles given in the following paragraphs:

a. LOCAL OSC.- A screw driver adjustment, equipped with a cap and captive chain assembly, shall be provided to permit manual tuning of the local oscillator

b. VOLUME.- A potentiometer control, fitted with a fluted knob and located on the test panel, shall be provided for increasing or decreasing the audio signal at the AURAL MON jack

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c. TEST OSC.- A screwdriver-operated control switch, located on the test panel and protected by a spring-loaded cover, with positions marked NORMAL, ON, TEST, shall be provided to permit (1) normal operation, (2) self-triggering of the beacon, and (3) suppression of extraneous pulses due to normal functioning of afc aiding circuits

d. RECEIVER GAIN.- A screwdriver-operated control, located on the test panel and protected by a spring-loaded cover, shall be provided to permit adjustment of the receiver gain

3.5.1.7 Indicator lamps.- A red indicator lamp marked TEST OSC and a white indicator lamp marked AFC shall be provided on the test panel.

3.5.1.8 Internal controls.- The following controls shall be placed on the assemblies indicated which shall be housed within the receiver-transmitter case and which shall be identified by the following titles:

a. LONG ADJ.- A screwdriver control, located on the pulse control chassis, shall be provided to permit pulse-length discriminator adjustment

b. SHORT ADJ.- A screwdriver control, located on the pulse control chassis, shall be provided to permit pulse-length discriminator adjustment

c. O.I. ADJ.- A screwdriver control, located on the pulse control chassis, shall be provided to permit over-interrogation adjustment

d. MAGNETRON MANUAL TUNING.- A momentary switch, located on the afc chassis, shall be provided to permit magnetron automatic frequency-control adjustment

e. GEN. ADJ.- A screwdriver control, located on the modulator and high-voltage power supply chassis, shall be provided to permit adjustment of the rate-generator feedback voltage

f. SCAN-AFC.- A slide switch, located on the afc chassis, shall be provided to permit magnetron automatic-frequency-control adjustment

g. BAL-AFC.- A slide switch, located on the afc chassis, shall be provided to permit local oscillator automatic-frequency-control adjustment

h. REFL ADJ.- A screwdriver control, located on the afc chassis, shall be provided to permit adjustment of the local-oscillator reflector voltage

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i. MOTOR BAL.- A screwdriver control, located on the afc chassis, shall be provided for magnetron automatic-frequency-control adjustment purposes

j. Plus 150 V ADJ.- A screwdriver control, located on the low-voltage power supply and coder chassis, shall be provided to permit adjustment of the regulated plus 150-volt supply

k. Plus 500 V to Plus 700 V ADJ.- A screwdriver control, located on the low-voltage power supply and coder chassis, shall be provided to permit adjustment of screen voltage applied to the modulator tube

l. ATTENUATOR, RED.- A screwdriver control, located on the microwave unit, shallbe provided to permit adjustment of local-oscillator power to the mixer

m. ATTENUATOR, GREEN.- A screwdriver control, located on the microwave unit, shall be provided to permit adjustment of the magnetron power to the 9308 mc discriminator cavity

n. ATTENUATOR, BLUE.- A screwdriver control, located on the microwave unit, shall be provided to permit adjustment of the magnetron power to the 9312 mc discriminator cavity

o. ATTENUATOR, YELLOW.- A screwdriver control, located on the microwave unit, shall be provided to permit adjustment of the local-oscillator power to the 9308 mc discriminator cavity

p. ATTENUATOR, WHITE.- A screwdriver control, located on the microwave unit, shall be provided to permit adjustment of the local-oscillator power to the 9312 mc discriminator cavity

3.5.1.9 Adjustment of internal controls.- Where practicable, internal controls shall be such that when the radio beacon is aligned and adjusted for optimum performance the positions of all controls shall be approximately in the center of their adjustment range.

3.5.1.10 Internal test jacks.- The following test points shall be available on assemblies indicated which shall be housed within the receiver-transmitter unit and which shall be identified by the following titles:

a. J-304. A pin jack, located on the pulse-control chassis, shall be provided as a test connection for checking the pulse-integrator output

b. J-305. A pin jack, located on the pulse-control chassis, shall be provided as a test connection for checking the pulse-length discriminator output

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c. J-306. A pin jack, located on the pulse-control chassis, shall be provided as a test connection for checking the blanking gate and suppressor-gate output

d. J-307. A pin jack, located on the pulse-control chassis, shall be provided as a test connection for checking the aural monitor and counter input

e. J-602, J-603. Two pin jacks, located on the afc chassis, shall be provided as test connections for checking the rate generator voltage

f. J-605. A BNC receptacle, type UG-290/U, located on the afc chassis, shall be provided as a test connection for checking the local-oscillator discriminator output voltage

g. J-607. A pin jack, located on the afc chassis, shall be provided as a test connection for checking the local-oscillator reflector voltage

h. J-609, J-610. Two pin jacks, located on the afc chassis, shall be provided as test connections for checking the magnetron tuning motor voltage

i. J-613, J-614. Two pin jacks, located on the afc chassis, shall be provided as test connections for checking the magnetron discriminator balance

3.5.1.11 Microwave unit

3.5.1.11.1 Composition.- The microwave unit shall provide a duplexer, mixer, local oscillator, magnetron, and discriminator consisting of directional couplers, waveguide attenuators, reference cavities and afc crystal detectors.

3.5.1.11.2 Function.- The microwave unit shall couple interrogation pulse energy from the antenna and transmission line system to the mixer crystal during reception, and magnetron pulse energy to the antenna during transmission. In addition, the microwave unit shall contain the components necessary to aid in maintaining the frequencies of the magnetron and the local oscillator at 9310 mcs.

3.5.1.11.3 Duplexer.- The duplexer shall use TR and ATR tubes located in such a manner as to provide maximum duplexing efficiency without adjustment by operating personnel. The TR and ATR tubes shall fire instantaneously so that the first relay pulse shall not damage the mixer crystal and shall not be attenuated as it is propagated to the antenna.

3.5.1.11.4 Mixer.- The mixer crystal shall be a JAN 1N23B silicon crystal. The crystal holder shall be designed for convenient removal and replacement of the crystal.

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3.5.1.11.5 Local oscillator.- The local oscillator shall be a JAN-2K25 reflex klystron. Provision for mechanically tuning the oscillator to proper frequency shall be provided. This shall be accomplished without removing the case of the receiver-transmitter. The output of the local oscillator shall be coupled to the mixer crystal through a 1 to 15 db adjustable attenuator and an 8.5 db directional coupler.

3.5.1.11.6 Discriminator.- The discriminator shall include two 24-db directional couplers and a 1 to 15 db adjustable attenuator placed between the magnetron and each reference cavity, and one 8.5-db directional coupler between the local oscillator and each reference cavity. A 3 to 10-db adjustable attenuator shall be placed between each reference cavity and its associated crystal detector. The two cyrstal detectors shall be type 1N23B. They shall detect the output of the reference cavities and shall supply the afc circuits with these rectified voltages. The crystal holders shall be an integral part of the microwave assembly and shall be designed for convenient removal or replacement of crystals. The reference cavities shall be JAN types 6041 and 6040, MIL-E-1/296.

3.5.1.11.7 Magnetron.- The transmitter tube shall be a tunable magnetron such as type 6233 in accordance with Specification MIL-T-8212, or equivalent, and shall be provided with a suitable driving motor and gear train to tune the magnetron.

3.5.1.12 Receiver circuit

3.5.1.12.1 Sensitivity, gain, and bandwidth.- The overall sensitivity and gain of the receiver over the bandwidth 9335 to 9415 megacycles shall be such that a 2.35 microsecond input signal of -66 dbm or less, measured at the input of the receiver-transmitter, shall provide sufficient receiver output to trigger the coder 100 percent of the time. Signal loss, if any, due to action of the pulse width discriminator circuits shall be included in the above stated system triggering sensitivity. The receiver shall not be so sensitive as to cause the beacon to transmit due to internal noise more than once per second averaged over a period of 1 minute.

3.5.1.13 Pulse control chassis

3.5.1.13.1 Pulse length discrimination.- A pulse length discrimination circuit shall be triggered by the receiver and shall in turn trigger the coder. The discriminator shall be so adjusted that it shall pass pulses of 2.35 plus or minus 0.15 microseconds duration with a receiver input of -66 dbm and greater, and shall provide output to fire the coder 100 percent of the time. It shall reject pulses shorter than 1.1 or longer than 5.0 microseconds when the receiver input level is 0.001 watt. This requirement shall apply at all frequencies in the specified receiver band. The discriminator circuit shall further be provided with adjustment controls to enable shift of the pulse width acceptance mid-value and rejection limits when so required.

3.5.1.13.2 Blanking.- A blanking circuit shall be incorporated to provide a blanking gate which shall insure freedom from beacon triggering during the period of transmission of the coded signal. This blanking period shall be 200 plus or minus 40 microseconds in duration.

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3.5.1.13.3 Suppressor circuit.- A suppressor circuit shall be provided to prevent the beacon from being triggered by a pulse from the aircraft's own radar. This suppression circuit shall be actuated by a pulse from the radar to render the beacon insensitive for a period of not less than 15 nor more than 25 microseconds. The suppression pulse shall be coincident with the radar pulse. The pulse shall be 0.3 microsecond to 20 microseconds in duration with a positive amplitude from 10 to 50 volts across an impedance of from 100 to 100,000 ohms resistance shunted by a 1500 micro-microfarad capacitance, and with a rise time from 10 to 90 percent of full amplitude in less than 0.3 microsecond.

3.5.1.13.4 Test oscillator.- A switching arrangement shall be provided to permit the transmitter to free-run (that is, to give an output signal without external triggering) in order that certain maintenance and preflight check procedures may be carried out. The switching arrangement shall also permit triggering the transmitter by positive video pulses of 3 to 20 microseconds duration and a peak value of 10 to 50 volts from the pulse generator connected to a suitable jack, marked "TEST TRIG," mounted on the front panel of the receiver-transmitter. The switching arrangement shall include provision to illuminate a red panel-mounted warning lamp when the switch is in other than the NORMAL position.

3.5.1.13.5 Duty cycle limiter.- An overload prevention system or duty cycle limiting circuit shall be provided to protect the transmitter from operating at duty cycles in excess of 0.3 percent. This overload protective system shall not discriminate against weak signals nor render the beacon completely inoperative.

3.5.1.13.6 AFC aiding circuits.- An afc aiding circuit shall be provided which shall enable the magnetron to transmit frequently enough to insure proper operation of the afc circuits at such time that interrogation signals are being received only at a random or intermittent rate insufficient to provide adequate afc error voltage. The afc aiding circuits shall cause the magnetron to transmit only single pulses, regardless of code selection, at a nominal 300 cps rate for a period of 30 seconds after the afc aiding circuits are energized. The afc aiding circuits shall be energized by 4 consecutive interrogating pulses at a 250cps or greater rate. The afc aiding circuits shall not be energized in the absence of interrogating signals or in the presence of solid or "searchlighted" interrogating. After the 30-second period of self transmission, which shall be determined by a timing motor, the afc aiding circuit shall return to a quiescent condition such that the cycle may be repeated after a nominal 90 second delay, if necessary, due to further intermittent interrogation. The beacon shall be capable of coded reply to proper interrogation during the afc aiding cycle.

3.5.1.13.6.1 Magnetron scan rate.- The magnetron tuning scan rate shall be such that the proper frequency shall be transmitted within 30 seconds after initial interrogation has activated the afc aiding circuit. A scan rate generator shall be mechanically coupled to the magnetron drive motor to provide inverse feedback or damping to the motor energizing circuits. The amount of feedback shall be controlled by a potentiometer adjustment. The minimum scan rate with the SCAN-AFC switch in the SCAN position shall be 3 megacycles per second per second.

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3.5.1.14 AFC chassis

3.5.1.14.1 Function.- The afc circuitry shall maintain operation of the magnetron and local oscillator at a frequency of 9310 ± 1-1/2 megacycles at temperatures above 0 degrees C and plus or minus 1.75 megacycles at temperatures between -55 degrees C and 0 degrees C by receiving outputs from the two afc detectors in the microwave discriminator and by providing corrective voltages to the magnetron tuning motor and local oscillator repeller. An adequate filtering circuit shall be included to separate pulsed error voltages and cw error voltages received from the discriminator due to drift of the magnetron and local oscillator from the desired frequency.

3.5.1.14.2 AFC range.- The afc circuits shall provide means for bringing the magnetron and local oscillator frequencies within the range of the discriminator circuits in the event their frequency deviation is greater than 2 megacycles off center frequency of 9310 megacycles.

3.5.1.14.3 Magnetron AFC.- The magnetron afc circuitry shall provide voltages capable of energizing the magnetron drive motor through the entire tuning range of the magnetron. Limit switches and a friction clutch shall be provided to insure reversal of the drive motor, at its tuning limits, and to prevent damage to the magnetron. The scan voltage shall be removed from the magnetron driving motor by a relay and switching tube when the magnetron frequency falls within the discriminator range. At such time the magnetron motor shall be controlled only by amplified error voltages originating from the reference cavities.

3.5.1.14.4 Local oscillator AFC.- The local oscillator afc circuitry shall include a means of varying the local oscillator frequency by applying a sine wave voltage to the repeller. This voltage shall be of sufficient amplitude to insure complete scan of the operating mode of the klystron in order that the klystron frequency will sweep into the range of the discriminator. As the local oscillator frequency is corrected and falls within the discriminator range, this SWEEP VOLTAGE shall be electronically removed from the local oscillator repeller. A panel mounted indicator lamp shall be energized during application of this SWEEP VOLTAGE. The local oscillator shall stabilize on frequency within 3 minutes after the equipment is energized and thereafter shall remain within the discriminator range until the radio beacon is de-energized.

3.5.1.14.5 False lock in.- The magnetron and local oscillator shall not LOCK on any frequency other than that specified in 3.5.1.14.1.

3.5.1.15 Low voltage power supply and coder chassis.

3.5.1.15.1 Low voltage power supply.- The low voltage power supply shall supply all low voltage circuits of the radio beacon with required filament and dc voltages.

3.5.1.15.2 Coder circuit.- The coder shall receive its input signal from the pulse length discriminator circuit and its output shall be used to key the transmitter.

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3.5.1.15.3 Type.- The coder shall include a magnetrostriction type delay-line coding device which shall provide, by selection, any combination of 1 to 9 pulses. The code must be free of jitter to the extent that each individual code element, when utilized, must be stable in time spacing within plus or minus 0.5 microsecond of the nominal position occupied by the respective code elements. The magnetrostriction line termination efficiency, under all service conditions, shall be such that reflections are nonexistent or of such low amplitude that false coder output cannot take place. The coder shall not limit the maximum reply rate capability of the beacon.

3.5.1.15.4 Code combination.- The number of pulses in any code combination will normally be a maximum of 6 and a minimum of 1. The initial code element, with respect to time, shall be present in all code combinations and it shall bear the time relation to the interrogating pulse specified in 3.5.1.18. Subsequent code elements shall be spaced at intervals of 6 plus or minus 1 microseconds from the preceding code element. However, tolerances shall not be cumulative in one direction and the interval between any two alternate code elements shall be 12 plus or minus 1 microseconds.

3.5.1.15.5 Code selection.- Code selection shall be remotely (electrically) controlled from Control, Radio Beacon C-812( )/APN-69.

3.5.1.16 Modulator and transmitter

3.5.1.16.1 Power output.- The peak power output of the transmitter into a matched load averaged over the duration of the pulse shall be at least 5000 watts. A matched load shall be considered as existing when the voltage standing wave ratio (VSWR) in the transmission line system is 1.5:1 or less.

3.5.1.16.2 Frequency stability.- The frequency of the transmitter, when connected to an antenna by a waveguide transmission line system which may vary in length from 2 to 30 feet and with a VSWR of 1.5:1, shall remain within plus or minus 1.5 megacycles of a nominal frequency of 9310 megacycles under the following conditions or natural combination thereof:

a. Change in duty cycle from 0.025 to 0.3 percent
b. All service conditions
c. Change in phase of the load of 150 degrees

3.5.1.16.3 Transmitted pulse characteristics.- The transmitted pulses shall be 0.5 plus or minus 0.1 microsecond in duration, measured at 90 percent peak amplitude, with a rise and decay time of less than 0.1 microsecond between 10 percent and 90 percent peak amplitude. The power deviation during the pulse width shall not exceed 1 db. Undesirable magnetron output following transmitted pulse shall be eliminated by providing a clipping diode to damp input circuit oscillation.

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3.5.1.16.4 Magnetron filament voltage relay.- A relay shall be provided to reduce the magnetron filament voltage from 6.3 volts to 4 plus or minus 1/2 volts when the duty cycle exceeds 0.2 percent.

3.5.1.17 Aural monitor.- Means shall be provided for aurally monitoring the beacon interrogation and its retransmission. Input to the monitor circuit shall be representative of the beacon interrogation when the equipment is in STANDBY condition, and shall be representative of the beacon transmission when the equipment is in OPERATE condition. Monitoring shall be accomplished by the use of a headset, cord and plug, and jack arrangement. The monitor output shall provide not less than 200 milliwatts power output to a nominal load impedance of 600 ohms.

3.5.1.18 Beacon delay.- The total delay of the beacon reply as measured from the trailing edge of the interrogating pulse to the leading edge of the first reply pulse shall be 1.5 plus or minus 0.5 microseconds.

3.5.1.19 Magnetron current metering.- The magnetron current shall be measured by a 0 to 1 milliampere meter connected to Jack J-502 located on the front panel of the receiver-transmitter. This jack shall be connected to a divider that has a 20:1 current step-down ratio. Thus, the meter shall indicate 1/20 of actual magnetron current. An additional divider shall be required to provide suitable deflection of a 0 to 3 milliampere meter which shall be an integral part of Test Set AN/APM-67. This divider shall be connected to Jack J-402, pin D.

3.5.1.20 Test points.- Maintenance test facilities shall be expanded to the greatest extent practicable through installation of chassis and panel mounted test point connectors electrically connected to selected portions of the various circuits to enable rapid derivation of circuit performance. The minimum number of test points acceptable shall be as indicated in 3.5.1.5.1 and 3.5.1.10.

3.5.1.20.1 Test set connection.- The contacts of the test set connector AN3102C-14S-5S designated as J-402, shall be utilized as follows:

Contact	USE
A	115 volt ac, 380 to 1000 cps
B	115 volt ac, 380 to 1000 cps
C	Crystal current metering
D	Magnetron current metering
E	Common metering return

3.5.1.20.2 Test trig.- A positive pulse of 10 to 50 volts amplitude and 1 to 10 microseconds duration shall be made available at the TEST TRIG connector for maintenance and adjustment purposes. This pulse shall be timed coincident with the pulse that triggers the coder.

3.5.1.21 RF radiation.- RF radiation due to the receiver local oscillator shall be less than 1 milliwatt when measured at the waveguide output at the front panel of the receiver-transmitter.

3.5.1.22 Primary power routing.- All power shall be brought directly to the receiver-transmitter and shall be applied to the unit through relays controlled by the OFF-STANDBY-OPERATE switch on Control C-812( )/APN-69.

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3.5.1.23 Fuses.- All primary power circuits of the receiver-transmitter shall be suitably protected by fuses or circuit breakers mounted on the test panel of the receiver-transmitter.

3.5.1.24 Power control circuits.- Suitable thermal time delay relays shall be employed in the primary power circuits of the plate transformers of the low voltage and the high voltage power supplies to prevent application of primary voltage prior to or coincident with application of filament voltage. The time delay relays shall permit energizing the low voltage plate transformer 30 seconds after filament voltage is applied, and energizing the high voltage plate transformer circuit 3 minutes after filament voltage is applied. Energizing the high voltage plate transformer, however, shall further require that the OFF-STANDBY OPERATE switch is in the OPERATE position. There shall be no further delay in energizing the high voltage plate transformer as the OFF-STANDBY-OPERATE switch is moved between STAND-BY and OPERATE positions, unless the equipment is previously completely de-energized.

3.5.1.24.1 Power-on indicator.- The high voltage transformer shall be provided with a low voltage tap suitable for energizing an indicator lamp as required by 3.5.2.7.

3.5.2 Control, Radio Beacon C-812( )/APN-69.- The Control, Radio Beacon C-812( )/APN-69 shall contain all switches, aural monitor volume level control and outlet, and circuits necessary for operational control of the radio beacon.

3.5.2.1 Size and shape.- The control shall be in accordance with Specification MIL-C-6781. The panel size and shape shall be in accordance with Drawing 53C12932. Orderly arrangement of switches without overcrowding to such an extent as to cause confusion or difficulty of manipulation or observance shall be provided.

3.5.2.2 Front panel controls

3.5.2.2.1 Power control switch.- A three-position rotary switch, actuated with a bar-knob type handle indicating switch position, for manually selecting OFF-STANDBY-OPERATE condition of the radio beacon shall be provided.

3.5.2.2.2 Aural gain control.- A potentiometer control, fitted with a fluted-knob and designated VOLUME, for increasing or decreasing the audio signal at the control unit aural monitor jack shall be provided.

3.5.2.2.3 Code selector.- Lift-to-slide type on-off switches shall be provided to permit control of the presence or absence of each reply code pulse element.

3.5.2.3 Code indication.- A green-jewel indicator shall be placed directly above a dummy-switch knob and shall be indicative of the initial reply pulse following interrogation. Eight additional and identical green-jewel indicators shall be placed directly above eight lift-to-slide type code selector switches. Illumination of a jewel shall indicate the presence of a code element. Both the position of the code-selector switches and their associated illuminated jewels shall indicate the code combination transmitted.

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3.5.2.3.1 Code selection.- A series of lift-to-slide type off-on switches to control the absence or presence of all available reply code pulse elements, except the initial reply pulse following interrogation, shall be provided. The switches shall be equipped with satisfactory mechanical means of providing readily observable indication of their electrical positions and shall, as a group, be designated CODE SELECTOR. A dummy switch and indicator in the ON position shall be provided in the space associated with the first reply pulse which shall be the initial pulse of any code.

3.5.2.4 Arrangement.- The code selection switches shall be arranged in a horizontal row across the front panel of the control in such a manner that their respective indicators, from left to right, are indicative of reply code pulse element presence or absence, from first to last, with respect to time.

3.5.2.5 Locking provision.- Code selection switches are intended as operational controls; however, means shall be provided to lock or secure the switches in any pre-determined combination for use during operational missions that may require a fixed reply code. The locking provision shall be such as to reduce the possibility of unintentional change of any switch in the code selection group to a minimum.

3.5.2.6 Front panel receptacle.- An aural monitor headset jack equipped with a spring-loaded cover shall be provided on the front panel in such location that the associated headset cord and plug shall not interfere with operational controls. The receptacle shall be designated AURAL MONITOR.

3.5.2.7 Indicator light.- A green-jewel indicator lamp, fitted with a dimmer and a push-to-test feature, shall be provided on the front panel of the control unit. The indicator lamp shall be energized from a suitable tap on the high voltage power supply transformer of the receiver-transmitter, thereby giving an indication when the radar beacon is capable of transmission. The indicator lamp shall be designated TRANSMITTER ON.

3.5.2.8 Electrical connections.- A multiple-contact receptacle, AN3102A-22-14P shall be provided for connection to the receiver-transmitter.

3.5.2.9 Power control.- The power control switch, designated OFF-STANDBY-OPERATE, shall control primary power input to the receiver-transmitter, by appropriate relay action, to accomplish the following:

a. OFF — Receiver-transmitter completely de-energized

b. STANDBY — All circuits of receiver-transmitter energized except the modulator high voltage supply

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c. OPERATE — All circuits of receiver-transmitter energized providing the switch has previously been in STANDBY or OPERATE positions for sufficient time to prevent equipment damage by applications of modulator voltage without sufficient warm-up time. The indicator light on the control panel shall be energized coincidently with application of modulator voltage.

In addition to the foregoing power switching function, the switch shall also cause the aural monitor circuit input to shift from INTERROGATION to TRANSMISSION when changing from STANDBY to OPERATE positions.

3.5.2.10 Aural monitor.- The aural monitor circuit jack and gain control provided at the panel of the receiver-transmitter in 3.5.1.17 shall be duplicated at the panel of Control, Radio Beacon C-812( )/APN-69. The control shall provide for increasing or decreasing the audio output at the monitor jack.

3.5.2.11 Code selector switches.- The code selector switches shall be positive acting off-on switches which shall control respective coding relays within the coder section of the receiver-transmitter.

3.5.3 Antenna AS-521( )/APN-69.- The antenna shall consist of a single radiating and receiving element, required transition sections to provide impedance matching and to enable coupling with standard waveguide fittings, and a suitable protective covering or radome. It shall be designed to be used with RT-204( )/APN-69 as described in 3.5.1.

3.5.3.1 Mechanical requirements.- The antenna and radome shall be capable of withstanding airspeeds up to 600 knots. The radome shall be designed to effect minimum drag, consistent with the electrical requirements, at the above speed. The radome shall be low-loss and shall resist impact, shock, vibration, and weather erosion. No degradation of mechanical nor electrical characteristics of the antenna shall result from immersion tests specified herein. The antenna and radome shall be capable of being pressurized. Pressure leakage rate shall not exceed 1/2 pound per square inch in 5 hours from a gage pressure of 20 pounds per square inch. Specification MIL-P-7094 shall be given consideration in choice of materials and processes. The dimensions of the antenna assembly shall not exceed the dimensions indicated on Drawing 53C12933.

3.5.3.1.1 Coupling.- The antenna assembly shall be terminated with a waveguide flange part MS90058-136A in accordance with Standard MS90058.

3.5.3.1.2 Mounting.- The antenna and radome shall be designed for mounting at the top of the fuselage of aircraft. The words DO NOT PAINT in red letters 1/2 inch high shall be stenciled on each side of the radome near the junction of the radome and antenna mounting. A nonconductive paint shall be used.

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3.5.3.2 Electrical requirements

3.5.3.2.1 Polarization.- The antenna shall be horizontally polarized.

3.5.3.2.2 Antenna radiation pattern.- The antenna, with radome in place, shall have the following free space characteristics when mounted on a 30-inch diameter ground plane. The pattern shall be omni-directional within 3.5 db in the horizontal plane. The elevation beamwidth at the half power points shall be 11.5 plus or minus 2 degrees. The elevation beam shall be centered within 1.5 degrees of the horizontal plane. The average azimuth gain shall be more than 7 db greater than a point source.

3.5.3.2.3 Standing wave ratio.- The VSWR of the antenna with radome in place shall not exceed 1.5:1 between 9335 and 9415 megacycles and shall not exceed 1.3:1 at 9310 megacycles.

3.5.3.2.4 Power.- The antenna shall be capable of radiating 8000 watts peak pulse power at 50,000 feet altitude.

3.5.4 Mounting MT-824( )/APN-69.- Two each Mounting MT-824( )/APN-69 shall provide a vibration isolated mounting for Receiver-Transmitter RT-204( )/APN-69.

3.5.4.1 Mechanical requirements.- The mounting shall consist of a primary member formed to an appropriate shape to accommodate the receiver-transmitter unit and shall be equipped with two all-metal vibration isolators of proper load range characteristics. Attachment of the load to the mounting shall be by captive winged thumb screws with provision for safety wiring. Dimensions shall not exceed those indicated in Drawing 53C12931.

3.5.4.2 Electrical requirements.- The metal case of the receiver-transmitter shall have a low-resistance electrical connection to the mounting surface, exclusive of the path through the vibration isolators, to permit grounding of the receiver-transmitter case to the structure of the aircraft.

3.5.5 Fan, Vaneaxial MD-135( )/APN-69

3.5.5.1 Composition.- An axial-type fan assembly designated Fan, Vaneaxial MD-135( )/APN-69, consisting of a motor and a propeller mounted in a protective housing shall be provided for cooling the receiver-transmitter.

3.5.5.2 Form factor.- The external fan shall be as small as is practicable, consistent with required performance and life, and shall not exceed the dimensions indicated in Drawing 53C12931.

3.5.5.3 Mounting requirements.- Provision for mounting the fan assembly to the receiver-transmitter container in three alternate positions shall be made in such a manner as to assure adequate and efficient cooling of the receiver-transmitter. In addition, an alternate means shall be provided for mounting the fan assembly to the structure of the aircraft near the receiver-transmitter.

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3.5.5.4 Lubrication.- The motor-driven fan assembly shall be permanently lubricated and shall not require lubrication maintenance throughout the life of the equipment.

3.5.5.5 Power requirements.- The motor-driven fan shall operate from a 115 volt, 380-1000 cps, ac power source. The power consumption shall be a minimum consistent with the functional requirements of the fan but shall not exceed 150 watts.

3.5.6 Coupler, Directional, CU-341( )/U.- The function of Coupler, Directional CU-341( )/U shall be to provide an rf test point, with known attenuation, for use in determination of operating characteristics of Receiver-Transmitter, Radio RT-204( )/APN-69.

3.5.6.1 Mechanical requirements.- The directional coupler shall be designed for installation in the waveguide transmission line system adjacent to the receiver-transmitter. It shall be arranged for permanent installation, and shall be of unidirectional type. The output fitting shall mate with Plug UG-21D/U, in accordance with Standard MS91236, and shall have a cap and captive chain. Dimensions shall not exceed those indicated on Drawing 53B12934. Correct orientation of the directional coupler shall be indicated by clearly marking ANT on both broad sides of the waveguide near the proper end. Both ends shall be fitted with UG-136A/U choke flanges, however, the flange that connects to the receiver-transmitter shall have 0.189 inch clearance holes in accordance with Drawing 53B12934. The directional coupler shall be capable of being pressurized.

3.5.6.2 Electrical requirements.- The coupling of the directional coupler shall be 20 plus or minus 0.8 db at 9310 mc and the directivity shall be at least 15 db. The actual coupling, within 0.1 db, shall be clearly marked on the device in a location near the type N output fitting and in such manner and location that the characteristic can be easily determined, when installed, by maintenance personnel.

3.6 Government-loaned property.- When the contract or purchase order so provides, the Government will loan the following to the contractor, upon his request

ITEM NO.	STOCK NO.	ITEM DESCRIPTION	SPEC OR DWG NO.	QUANTITY
1		Beacon, Radio AN/APN-69	53C12931	1

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- The requirements of section 4 of Specification MIL-E-5400 and the following are applicable to this specification.

4.1.1 Design data.- The design data required in Specification MIL-E-5400 shall be furnished and shall include a cabling diagram showing only primary power circuits. The contractor shall furnish in addition a table showing circuit voltage, maximum currents and allowable voltage drops in each conductor of interconnection cables.

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4.2 Test conditions.- Unless otherwise specified, the equipment shall be subjected to the tests under the conditions specified in Specification MIL-E-5400 using the applicable reference voltages specified in Specification MIL-E-7894.

4.3 Preproduction tests

4.3.1 Sampling instructions.- The contractor shall conduct tests on three sample equipments to determine that the design of the equipments as proposed by the contractor will meet the requirements of all applicable specifications. The contractor's tests shall be conducted in accordance with the preproduction test procedures and in the order listed herein. The data obtained by the contractor in conducting thest tests shall be included with the design data requirements of Specification MIL-E-5400. The procuring activity shall be given at least 10 days prior notice when these tests are to be conducted in order that they may be supervised or witnessed, if desired, by a Government representative. The acceptance of the samples shall be contingent upon approval of all nonstandard components by the procuring activity.

4.3.1.1 Testing schedule.- The three sample equipments shall be processed as follows:

a. Sample No. 1: Sample No. 1 shall be subjected to the following tests:

(1) Electrical tests as specified in 4.3.2.1
(2) Service condition tests as follows:

(a) Temperature-altitude
(b) Humidity as specified in 4.3.2.2.3
(c) Vibration
(d) Interference

b. Sample No. 2: Sample No. 2 shall be subjected to the following tests:

(1) Electrical tests as specified in 4.3.2.1

c. Sample No. 3: Sample No. 3 shall be subjected to the following tests:

(1) Electrical Tests as specified in 4.3.2.1
(2) Those preproduction tests specified herein which were not performed on Sample No. 1. Included in the tests performed on Sample No. 3 shall be the humidity test specified in 4.3.2.2.3.1

4.3.2 Preproduction test procedures.- The preproduction test shall consist of the following tests and those specified under the acceptance test methods, subject to the test conditions specified herein. The electrical tests in 4.3.2.1 shall be performed prior to the service conditions tests.

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4.3.2.1 Electrical tests.- The following characteristics of the radio beacon shall be measured:

a. Transmitter frequency over specified duty cycle range, utilizing minimum to maximum number of code elements and related interrogation repetitic rate frequencies

b. Transmitter frequency over specified load phase change

c. Transmitter pulse power output over specified duty cycle range, utilizing minimum and maximum number of code elements. Pulse width shall be determined and pulse power computed from average power and duty cycle

d. Transmitted pulse spectrum utilizing minimum to maximum number of code elements, with attention to display at instants of duty cycle change

e. Transmitter tuning scan cycle duration in seconds and excursion in mc/sec., determination of proper operation of scan reversing mechanism, and stability of sense and positive lock-on of tuning mechanism incident with initial approach to 9310 megacycles

f. Transmitter tuning scan rate in megacycles per second per second when outside discriminator limits

g. Symmetry, balance and sense of magnetron afc circuits as evidenced by relationship of magnetron tuning motor voltage to frequency deviation from 9310 megacycles, within discriminator limits

h. System triggering sensitivity with 2.35 microseconds input signal, between limits of 9335 megacycles and 9415 megacycles

i. Pulse length discriminator acceptance range with pulse input signal of plus 10 dbm to -66 dbm

j. Over-interrogation control operation; relationship of magnetron current to duty cycle

k. Code spacing intervals and stability of code elements in all usable code combinations

l. Total beacon delay, with respect to the trailing edge of the interrogation pulse, with input signal levels of -30 dbm and -66 dbm

m. Proper operation of suppressor circuit as evidenced by decommission of the beacon for the specified interval following application of an external suppressor pulse

n. Proper operation of blanking gate circuits as evidenced by the beacon's nonsusceptibility to interrogation for the specified interval immediately subsequent to a successful interrogation

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o. Proper operation of aural monitor circuit, including smooth operation of both volume controls, as evidenced by the monitor tone being representative of successful interrogation when the beacon is in STANDBY condition, and the monitor tone being generated by the modulator when the beacon is in OPERATE condition

p. Proper functioning of magnetron afc aiding circuits as evidenced by:

(1) Quiescent condition in the absence of interrogation
(2) Initiation of cycle with four or more interrogation pulses at 250 cps, or greater, repetition frequency
(3) Determination of cycle duration
(4) Single pulse transmission during active portion of cycle
(5) Return to quiescent condition under either continuous interrogation or complete absence of interrogation

q. Proper operation of test oscillator or self-triggering function, with test oscillator switch in ON position, as evidenced by:

(1) Pulse repetition frequency
(2) Transmission of coded signal
(3) Illumination of panel warning indicator lamp
(4) Synchronizing signal available at panel SYNC TRIG receptacle

r. Proper operation of test oscillator circuit function, with test oscillator switch in TEST position, as evidenced by:

(1) Absence of self triggering from either test oscillator or afc aid circuits and susceptibility to triggering from signal as applied to panel TEST TRIG receptacle

(2) Transmission of coded signal

(3) Illumination of panel warning indicator lamp

s. Proper functioning of local oscillator repeller tuning circuits, when outside of discriminator control limits, as evidenced by sufficient scan voltage to adequately cover the operating mode under all operating conditions and thereby providing means of bringing the local oscillator frequency to such value that the discriminator may provide control, and subsequent maintenance of design operating frequency within the discriminator control limits. Periods during which the scan voltage is being applied to the local oscillator repeller will be determined to be coincident with illumination of the associated panel warning indicator lamp

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t. Proper reduction of magnetron filament voltage when duty cycle exceeds 0.2 percent

u. Temperature rise within Receiver-Transmitter, Radio RT-204( )/APN-69, and time required to reach stabilization

v. Proper functioning of Control, Radio Beacon C-812( )/APN-69, including intensity, uniformity, and contrast of panel lighting

w. Antenna radiation patterns in horizontal, longitudinal vertical, and transverse vertical planes

4.3.2.2 Service conditions (environmental)

4.3.2.2.1 Chamber open.- Before the start of the temperature cycle, the following characteristics of the beacon shall be determined and recorded in terms of numerical quantities, if applicable, otherwise in terms of satisfactory or unsatisfactory performance as per requirements of this specification.

a. Transmitter frequency-duty cycle relation

b. Transmitter pulse width, and power - - duty cycle relation

c. Transmitter spectrum

d. Proper operation of transmitter frequency scan cycle and lock-on

e. System triggering sensitivity with 2.35 microseconds input signal at 9335, 9375 and 9415 megacycles

f. Pulse length discriminator acceptance range with pulse input signal of -30 dbm and -66 dbm

g. Proper operation of over-interrogation control including magnetron current - - duty cycle relation

h. Proper code spacing intervals and stability of code elements

i. Total beacon delay with respect to the trailing edge of the interrogation pulse with input signal levels of -30 dbm and -66 dbm

j. Proper operation of suppressor circuit function

k. Proper operation of blanking gate and aural monitor functions

l. Proper operation of self triggering function with test oscillator switch in ON position, and in TEST position

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m. Proper operation of afc aiding circuit function

n. Proper operation of local oscillator afc function

o. Operation of thermostatic blower controls

p. Temperature rise within RT-204( )/APN-69

q. Proper operation of Control C-812( )/APN-69

4.3.2.2.2 Temperature-altitude.- The Temperature-Altitude Test Procedure I, including step 3 thereof, of Specification MIL-E-5272 shall be conducted for one cycle only. The rates of temperature and pressure change shall be as high as chamber construction permits except that the rates need not exceed those specified in Specification MIL-E-5400. The change from room temperature to -62 degrees C shall not require more than 4 hours.

4.3.2.2.2.1 With the radio beacon energized, the chamber pressure shall be returned to normal. A thorough visual inspection of each major component shall be made to determine that no damaging arcs were caused by operation at reduced pressure and also to determine that no normally sealed components show leakage as a result of operation at reduced pressure. After it has been determined visually that no damage has resulted from the temperature-altitude test, the tests specified in 4.3.2.2.1, with the exception of part p thereof, shall be repeated.

4.3.2.2.2.2 Thermocouples.- The following temperatures shall be measured by the thermocouple method during the temperature testing cycle:

No. 1 Temperature at a point inside each major component dust cover, the point to be chosen to represent as nearly as possible the average ambient temperature inside that dust cover

No. 2 Contact temperature on the largest transformer or inductor (except radio frequency inductors) or other large centrally located mass in each major component

No. 3 Contact temperature on the component or components where the highest operating temperature is expected, including tubes

4.3.2.2.3 Humidity.- A test shall be conducted in accordance with Specification MIL-E-5272, Humidity Test, Procedure I, for 12 days except that the maximum temperature shall be 50 degrees C. The test specified in 4.3.2.2.1, with the exception of part p thereof, shall be repeated once each day when the temperature is 38 degrees C or less. The equipment shall be operating during the test period only.

4.3.2.2.3.1 The humidity test to be performed on Sample 1 as required in 4.3.1.1 shall be the same as described in 4.3.2.2.3, except the duration shall be 48 hours.

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4.3.2.2.4 Pressure test

4.3.2.2.4.1 Receiver-transmitter.- The receiver-transmitter shall be subjected to a pressure test of 20 pounds per square inch gage pressure at temperatures of -55 degrees C, plus 25 degrees C, and plus 71 degrees C, and the pressure drop over a 24-hour period determined.

4.3.2.2.4.2 Antenna.- The antenna shall be subjected to a pressure test of 20 pounds per square inch gage pressure at normal room ambient conditions and the pressure drop over a 48 hour period determined.

4.3.2.2.5 Antenna immersion test.- The antenna radome shall be subjected to Immersion Test Procedure I as specified in Specification MIL-E-5272. The antenna shall then be tested in accordance with 4.3.2.1.

4.3.2.3 Service conditions (mechanical)

4.3.2.3.1 Vibration.- A vibration test shall be made in accordance with Specification MIL-E-5272, vibration test, procedure I. In addition, that portion of test procedure XI requiring that vibration isolators be removed shall be conducted for those equipments normally mounted on vibration isolators. The equipment shall be nonoperating during the procedure XI portion of the test. Satisfactory operation of the radio beacon shall be determined by repeating the tests specified in 4.3.2.2.1 except parts i, j, l, o, and p thereof, at intervals during this test (at least once in each plane of vibration). At the termination of the vibration test, a thorough visual inspection of each major component shall be made to detect any damage caused by vibration. After the above visual inspection, the tests specified in 4.3.2.2.1, with the exception of part p thereof, shall be repeated to determine that there has been no degradation in the performance of the radio beacon below the requirements of this specification as a result of the vibration test.

4.3.2.3.2 Shock.- A shock test shall be made in accordance with Specification MIL-E-5272, shock test, procedure II. The equipment shall be nonoperating during this test. At the termination of the test a thorough visual inspection of each major component shall be made to detect any damage. After the visual inspection the tests specified in 4.3.2.2.1, with the exception of part p thereof, shall be repeated to determine that there has been no degradation in the performance of the radio beacon below the requirements of this specification as a result of the shock tests.

4.3.2.4 Service conditions (electrical).- The radio beacon shall be subjected to tests that will demonstrate compliance with power supply requirements of this specification. These tests shall include tests as outlined in 4.3.2.1 excepting w, with the single-phase and dc power sources adjusted to extreme limits of specified voltage and frequency.

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4.3.2.4.1 Primary power.- Compliance with the paragraph titled Power requirements shall be ascertained concurrently with the following power factor test.

4.3.2.4.1.1 Power factor.- Compliance with equipment power factor requirements of Specification MIL-E-7894 shall be ascertained, using the following instruments:

Wattmeter - Weston Model 310, Form 3, or approved equal
Voltmeter - Weston Model 341, or approved equal
Ammeter - Weston Model 370, or approved equal

The sign of the power factor shall be determined by connecting a suitable capacitor across the power input while the equipment is in operation, and measuring its effect on the resultant power factor.

4.3.2.4.2 Interference.- Compliance with electrical interference requirements shall be checked in accordance with Specification MIL-I-6181.

4.3.2.4.3 Power source transient response.- The equipment shall be tested for transient response in accordance with figures 1 and 2.

4.3.2.5 Service conditions (special)

4.3.2.5.1 Sand and dust.- A sand and dust test shall be made in accordance with Specification MIL-E-5272, sand and dust test procedure I. After cooling to room temperature, the radio beacon shall meet the tests specified in 4.3.2.2.1 with the exception of part p thereof. A thorough visual inspection of each major component shall be made to determine that no damage has resulted from arcing or from mechanical wear.

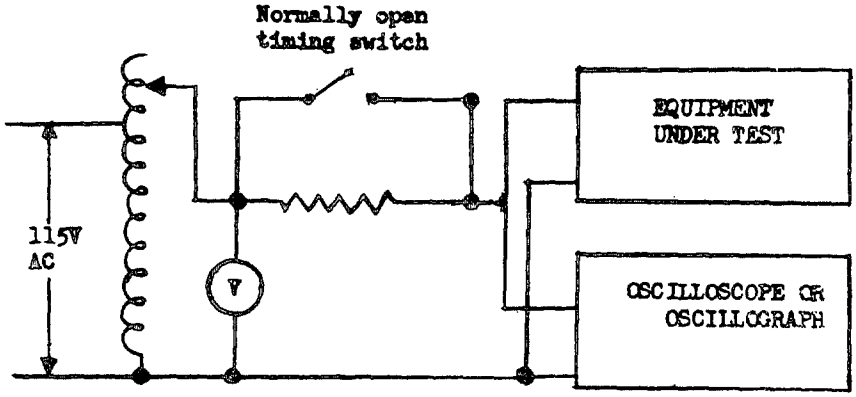
4.3.2.5.2 Equipment stability.- All of the components required to make up a complete Radio Beacon AN/APN-69 system shall be put into operation with all dust covers and shock mountings in place and with suitable test equipment connected to RF transmission line. After the equipment has operated for an initial 15-minute warm-up period, it shall be adjusted for optimum performance. The beacon shall then be allowed to operate continuously for a period of 48 hours without any further adjustment whatsoever. Changes in receiver sensitivity, pulse width acceptance range, power output, transmitter frequency, and code spacing that occur during the 48-hour test shall be noted in order to determine compliance with the paragraph titled STABILITY. There shall be no significant changes in the performance throughout this test.

4.3.2.5.3 Explosion proof.- The equipment shall be tested for explosion proofness in accordance with Specification MIL-E-5272, explosion proof test, procedure I.

4.3.2.5.4 Life.- The equipment shall be subjected to the 1000-hour life test under the following service conditions:

Temperature	Above 35 degrees C
Altitude	Normal (0 to 5000 feet)

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The power source rating in kva should be considerably greater than the load kva in order to avoid difficulty in the determination of the effective voltage during the transient. Each of the three transients shall be repeated at least three times. The interval between transients shall not be less than 30 seconds. Tolerances shall be ±2% on voltages and ±10% on time intervals

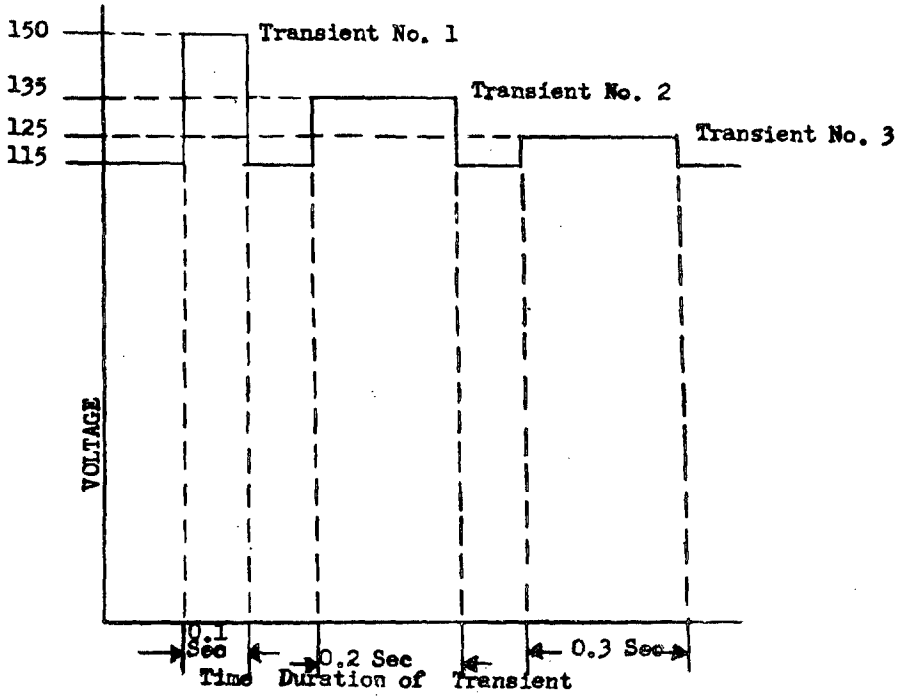
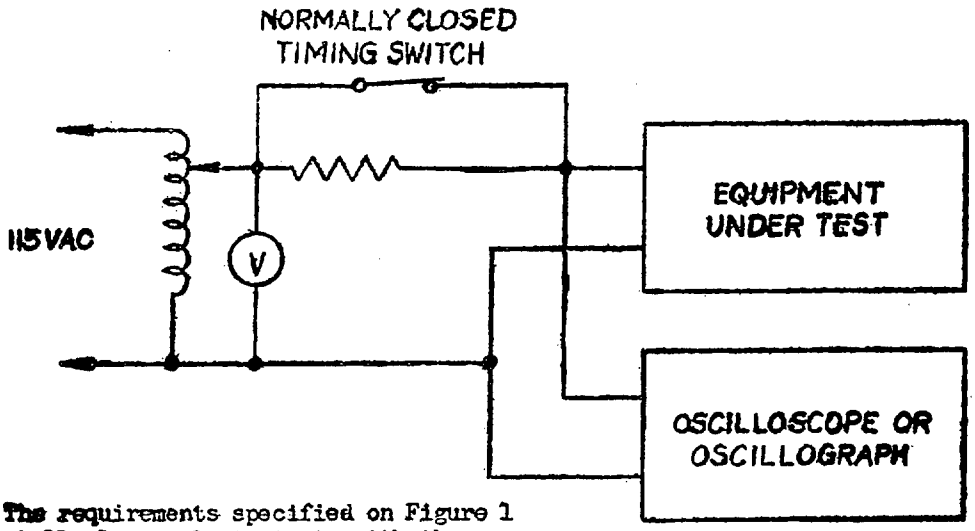


FIGURE 1. Overvoltage Transient Tests

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The requirements specified on Figure 1 shall also apply to tests with these low voltage transients.

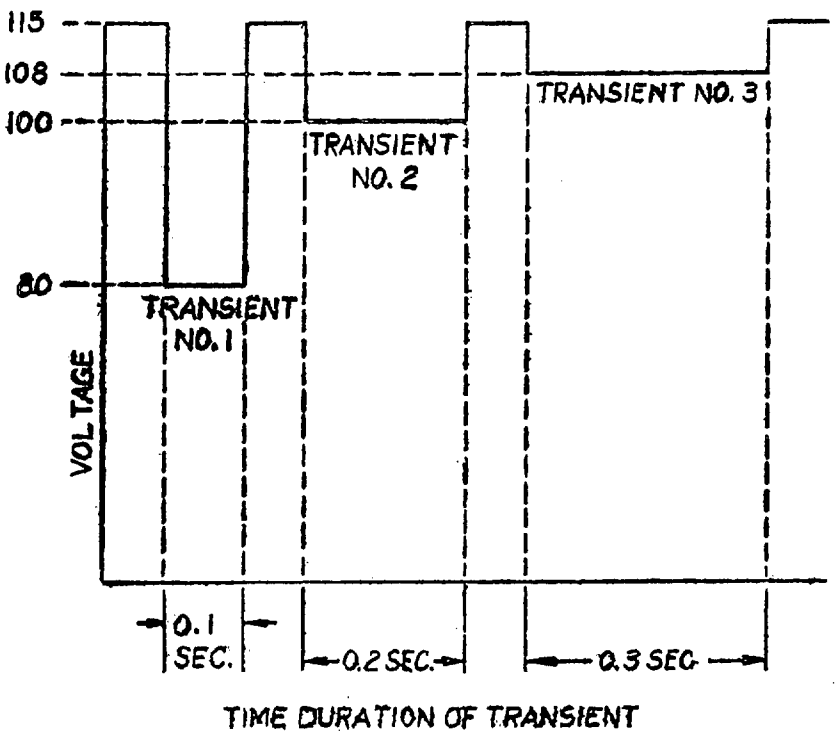


FIGURE 2. Undervoltage Transient Tests

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Humidity	Room ambient (up to 90 percent)
Ac voltage	115 to 120 volts (at lowest applicable frequency)
Dc voltage	27 to 29 volts
Vibration	Intermittent operation, constant frequency

A daily record as to the performance of the equipment, making particular note of any deficiencies shall be maintained. In the event of parts failures during this test the defective parts shall be replaced and the operation resumed for the balance of the 1000 hour test period. A record shall be kept of all such failures throughout the test indicating (a) the part type number, (b) the circuit reference symbol number, (c) the part function, (d) the name of the manufacturer, (e) the nature of the failure, (f) the number of hours the part operated prior to failure. A report shall be submitted to the procuring activity immediately upon completion of the first 200 hours of operation. In this report the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity shall review such proposals and determine whether they are acceptable. In addition, at the completion of the 1000-hour test, the contractor shall submit to the procuring activity similar report of all parts failures during the last 800-hours of test.

4.3.2.5.4.1 Once in each 24 hours of operation of the equipment it shall be subjected to a 10-minute period of vibration. The vibration shall be constant frequency, within the range of 25 to 35 cps, and total excursion of 0.015 to 0.030 inches, and need not be simple harmonic or purely linear motion. The operation of the equipment shall be observed during the 10-minute vibration period to determine whether degradation has occurred due to microphonics, worn or damaged parts, charred insulation, et cetera. After the 10-minute vibration period, the tests specified in 4.3.2.2.1, with the exception of part p thereof, shall be made. This cycle shall be repeated until the equipment has operated for the time specified herein.

4.4 Acceptance tests.- Acceptance tests shall consist of individual and sampling tests.

4.4.1 Individual tests.- Each radio beacon shall be subjected to the following tests.

4.4.1.1 Mechanical.- A visual inspection of the radio beacon shall be made to verify that the materials, design and construction, necessary mechanical measurements, marking and workmanship comply with the requirements of the specification.

4.4.1.2 Electrical.- The procedure for individual tests on the radio beacon, its components, assemblies, and sub-assemblies, shall be prepared by the contractor and shall be subject to the approval of the procuring activity. When approved, these tests shall form a part of this specification.

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4.4.2 Sampling tests.- Radio beacons shall be selected at random in accordance with the following:

QUANTITY OF ITEM ON ORDER	QUANTITY OF ITEMS TO BE TESTED
1 to 25	1
26 to 50	2
51 to 75	3
76 to 125	4
126to 200	5
201 to 300	6
Over 300	One for each additional 100 on order

4.4.2.1 Sampling tests shall consist of the tests specified in 4.3 except sand and dust and explosion proof tests to the extent considered necessary by the procuring activity. The duration of the life test shall be 500 hours.

4.4.2.2 When sampling tests are specified on a number of items that are selected from a production run and one or more of this number fails to meet the specified tests, acceptance of all items still on hand and subsequent production will be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of sampling test failure. However, final acceptance of the items on hand and subsequent production is contingent upon the inspector's decision regarding the over-all conformance of the product to specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that the defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for correction, will be furnished to the procuring activity.

4.5 Test procedures.- If the procedures and methods for conducting the tests are not stated herein, they shall be prepared by the contractor and sent to the procuring activity for approval. The right is reserved by the procuring activity to require additional tests when this is deemed necessary to determine compliance with the requirements specified herein.

5. PREPARATION FOR DELIVERY

5.1 General.- Preparation for delivery shall be in accordance with Specification MIL-E-17555.

5.1.1 Marking of shipment.- Internal packages and external shipping containers shall be marked in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Intended use.- The radio beacon covered by this specification is intended for use in aircraft and operates in a frequency range from 9335 megacycles to 9415 megacycles for interrogation, and 9310 megacycles for transmission.

6.2 Ordering data.- Invitation for bids, contracts, and orders should state the condition for the following:

6.2.1 Government model.- A standard sample of Beacon, Radio AN/APN-69 and drawings thereof should be made available for inspection by prospective bidders at a point to be designated in the request for proposal.

6.2.2 Preproduction tests.- It is expected that the contract or purchase order will specify that three radio beacons will be required as preproduction samples and that these preproduction samples will be subjected to the preproduction tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests. The time of delivery of each sample and its test data should be specified as concurrent with the time of the completion of its tests as listed in testing schedules in section 4.

6.2.3 Additional component parts.- Sub component parts, as specified in this specification, which are subject to approval by the procuring activity should be furnished in addition to the quantities of items ordered.

6.2.4 Accessory material and design data.- Procurement of ACCESSORY MATERIAL and DESIGN DATA as specified in Specification MIL-E-5400 should be furnished.

6.2.5 Other parts.- Such other parts as are necessary for the accomplishment of the preproduction tests should be furnished.

6.3 Reliability.- It is suggested that the reliability design procedure formulated jointly by Armed Services Electro Standards Agency and Radio-Television Manufacturers Association be consulted as a general guide. This check list is RTMA Chart No. 1. available from RTMA Engineering Department, 489 Fifth Avenue, New York 17, New York.

6.4 Life test set-up.- It is not intended that costly vibration testing machines be used for performing the 10-minute intermittent vibration during the life test as specified herein. A suitable platform may be mounted on vibration isolators, and a small motor with adjustable eccentric weights, or other simple vibrating device, may be used to shake this platform with the sample equipment mounted thereon. The test-set up shall be subject to the approval of the procuring activity.

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